

helper

JSON Merger

User Manual

Merge multiple Jira JSON files into one

1. What is helper?

The **helper** module is a utility tool that merges multiple Jira REST API JSON files into a single file. It is needed when a Jira export was split across multiple files due to API limits and **transform_data** expects a single file.

helper automatically deduplicates issues by their unique issue ID and generates a correct Jira API envelope (startAt, total, maxResults). The output file can be processed directly by transform_data.

2. Starting

2.1 As GUI (graphical interface)

Double-click the appropriate launcher in the extracted folder:

Operating system	Start file
Windows	Helper.bat (double-click)
macOS	Helper.command (right-click -> Open)
Linux	./Helper.sh

2.2 From source code

```
python -m helper
```

3. GUI Usage

The main window consists of three areas: the file list (top), the output file (middle), and the log area (bottom).

3.1 Adding input files

In the **Input files** area, one or more Jira JSON files can be added:

- **Add...** — Opens a file dialog where multiple files can be selected at once (Ctrl+click or Shift+click).
- **Remove** — Deletes the selected entries from the list (the files on disk are not affected).

Tip: The order in the list does not affect the result — all issues are merged and deduplicated independently of file order.

3.2 Choosing the output file

Enter the output file path in the **Output file (JSON)** field or select it using the **Browse...** button. The file is created when merging (overwritten if it already exists).

3.3 Remove duplicates

The checkbox **Remove duplicates (by issue id)** is enabled by default. Issues appearing in multiple input files with the same ID are written to the output only once. Each removed duplicate is logged.

Note: Deduplication is based on the **id** field in the Jira JSON (numeric issue ID). The **key** field (e.g. ARTA-123) is used for log display only.

3.4 Starting the merge

Click the **Merge** button. Progress is shown in the log area. For longer operations a progress bar appears. After completion the log shows the number of merged and deduplicated issues along with the output file path.

4. CLI Mode

helper can also be called directly from the command line without a GUI:

```
python -m helper file1.json file2.json --output merged.json
```

Parameter	Description
FILE ...	At least one input JSON file (required)
--output FILE	Output file path (required)
--no-dedup	Disable deduplication (keep all issues including duplicates)

5. Output Format

The output file is a valid Jira REST API JSON in the format of Jira Cloud API v2 / v3. The envelope is set automatically:

Field	Value
expand	Taken from the first input file
startAt	Always 0
maxResults	Number of merged issues
total	Number of merged issues
issues	All issues from all input files (after dedup)

Compatibility: The output file can be used directly as input for **transform_data** — no further processing required.

6. Frequently Asked Questions (FAQ)

Q: How many files can I merge?

A: There is no technical limit. In practice 2-10 files are typical (corresponding to 200-10,000 issues depending on pagination size).

Q: What happens if an input file has no 'issues' field?

A: helper aborts with an error message and writes no output file. Check that the file is a valid Jira REST API response.

Q: Can I load the output directly into transform_data?

A: Yes. The output file contains the same envelope as a regular Jira API response and is accepted by transform_data.

Q: What does 'duplicate' mean in this context?

A: An issue is a duplicate if two entries share the same numeric 'id'. This can happen when the same issue was returned in multiple paginated API queries.

Q: Can I use JSON files from Jira Cloud API v3?

A: Yes. helper supports both Jira API v2 and v3 — the 'issues' key is present in both versions.

7. Glossary

Term	Explanation
Deduplication	Removing duplicate issues based on the numeric issue ID.
Envelope	The outer wrapper of the Jira API JSON response containing fields like startAt, total, and maxResults.
Issue ID	Unique numeric ID of a Jira issue (field 'id'). Different from the issue key (e.g. ARTA-123).
Jira Cloud API	REST API of Jira Cloud; returns issues as JSON with an 'issues' array.
JSON	JavaScript Object Notation — lightweight text format for structured data.
Pagination	Splitting large API results across multiple requests (pages) due to API limits.
transform_data	SituationReport module that converts Jira JSON data into an XLSX file (IssueTimes).